

TEST 1

1. What is the difference between a fundamental and derived unit? Give an example of each. (2)
2. Write the following in scientific notation:
(a) 240500 (b) 0.00419 (c) 6439×10^2 (d) 21.4×10^{-4} (4)
3. Convert the following to standard units. (Express your answers in scientific notation.)
(a) 346 km (b) 2.45 mm^2 (c) $2.39 \times 10^{-4} \text{ g}$ (d) $579 \mu\text{s}$ (4)
4. How many significant figures are in the following:
(a) 6410 (b) 0.0190 (c) 101.3 (d) 2.00×10^4 (4)
5. Perform the following calculations, giving the answers to the appropriate number of significant figures.
(a) $216.4 + 41.02 + 62$

(b) $0.0416 + 1.24 - 0.12$

(c)
$$\frac{(2.29 \times 10^3)(1.004 \times 10^{-2})}{(6.2 \times 10^3)}$$

(d)
$$\frac{(24.9 + 6.42 + 19.451)}{(1.03 \times 10^2)}$$
 (6)

TOTAL: 20 MARKS

YEAR 10 PHYSICS
TEST 1

W-90

1. FUNDAMENTAL UNIT: used to express all other units/quantities e.g. kg, m, s.

DERIVED UNIT: based on the fundamental units e.g. ms^{-1} , ms^{-2} , kg ms^{-1} . (2)

2. (a) 2.405×10^5 (b) 4.19×10^{-3} (c) 6.439×10^5 (d) 2.14×10^{-3} (4)

3. (a) $3.46 \times 10^5 \text{ m}$ (b) $2.45 \times 10^{-6} \text{ m}^2$ (c) $2.39 \times 10^{-7} \text{ kg}$ (d) $5.79 \times 10^{-4} \text{ s}$ (4)

4. (a) 3 (b) 3 (c) 4 (d) 3 (4)

5. (a) $\begin{array}{r} 216.4 \\ 41.02 \\ + 62 \\ \hline 319.42 \end{array}$	(b) $\begin{array}{r} 0.0416 \\ + 1.24 \\ - 0.12 \\ \hline 1.1616 \end{array}$
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$\therefore \text{ans.} = 3.19 \times 10^2$ (1½)

$\therefore \text{ans.} = 1.16$ (1½)

(c)
$$\frac{(2.29 \times 10^3)(1.004 \times 10^{-2})}{(6.2 \times 10^3)}$$

$= 3.708 \times 10^{-3}$

(d)
$$\frac{(24.9 + 6.42 + 19.451)}{(1.03 \times 10^2)}$$

$= \frac{(50.8)}{(1.03 \times 10^2)}$

$\therefore \text{ans.} = 3.7 \times 10^{-3}$ (1½)

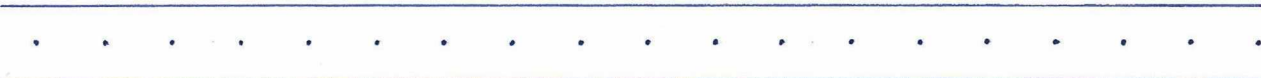
$= 4.932 \times 10^{-1}$

$\therefore \text{ans.} = 0.493$ (1½)

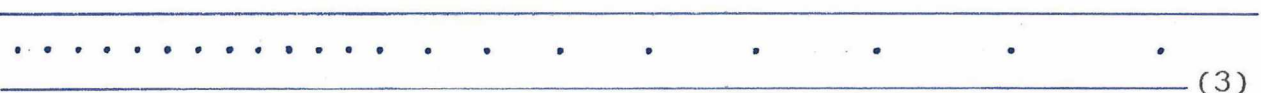
TOTAL: 20.

TEST 2

1. Describe the motion shown in the following ticker tapes.

(a) 

(b) 

(c)  (3)

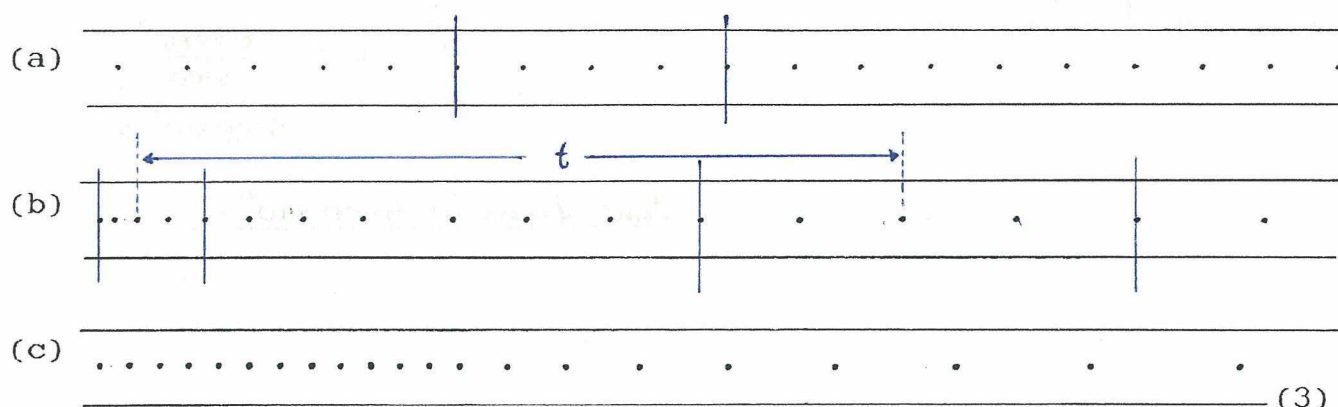
2. For the tapes (a) and (b) in Question 1, calculate the average velocity and the acceleration where appropriate. Set your working out clearly. (5)

3. A runner moving at a constant velocity of 5.00 ms^{-1} covers a distance of 2.50 km. How long did it take to cover this distance? (3)

TOTAL: 11 MARKS

TEST 2

1. Describe the motion shown in the following ticker tapes.



(3)

2. For the tapes (a) and (b) in Question 1, calculate the average velocity and the acceleration where appropriate. Set your working out clearly. (5)

3. A runner moving at a constant velocity of 5.00 ms^{-1} covers a distance of 2.50 km. How long did it take to cover this distance? (3)

TOTAL: 11 MARKS

ANSWERS.

1. (a) CONSTANT VELOCITY
 (b) CONSTANT ACCELERATION
 (c) CONSTANT VELOCITY, THEN CONSTANT ACCELERATION (3)

2. (a)
$$v = \frac{s}{t}$$

$$= \frac{3.60 \times 10^{-2}}{0.0800}$$

$$= 0.450 \text{ ms}^{-1}$$

$$\therefore \text{constant velocity is } 0.450 \text{ ms}^{-1} \quad (2)$$

(b)
$$u = \frac{s}{t}$$

$$= \frac{1.40 \times 10^{-2}}{0.0800}$$

$$= 0.175 \text{ ms}^{-1}$$

$$v = \frac{s}{t}$$

$$= \frac{5.80 \times 10^{-2}}{0.0800}$$

$$= 0.725 \text{ s}$$

$$t = 11 \text{ dot spacings} = 0.220 \text{ s}$$

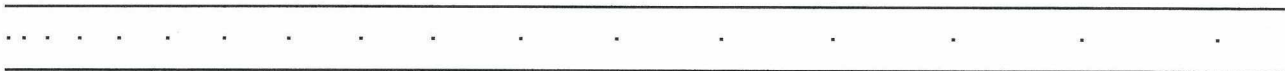
$$a = \frac{v-u}{t}$$

$$= \frac{0.725 - 0.175}{0.220}$$

$$= 2.50 \text{ ms}^{-2} \quad (3)$$

YEAR 10 SCIENCE
PHYSICS
TEST 2

1. During an orienteering event, an athlete moved in the following way: 2.00×10^2 m W in 50.0 s, 4.50×10^2 m NE in 2.00 minutes and 3.00×10^2 m S in 75.0 s. Calculate:
- (a) the athlete's displacement.
 - (b) the average velocity during the movement.
 - (c) the average speed for the athlete. (5)
2. The ticker tape below was produced on a machine that operates on 50 dots per second.



Analyse the tape and determine the acceleration represented on it. (4)

3. A car moving at 20.0 ms^{-1} along a flat road brakes uniformly for 3.00 s and reduces its velocity to 5.00 ms^{-1} . Calculate:
- (a) the acceleration of the car.
 - (b) the distance covered by the car during this time. (4)

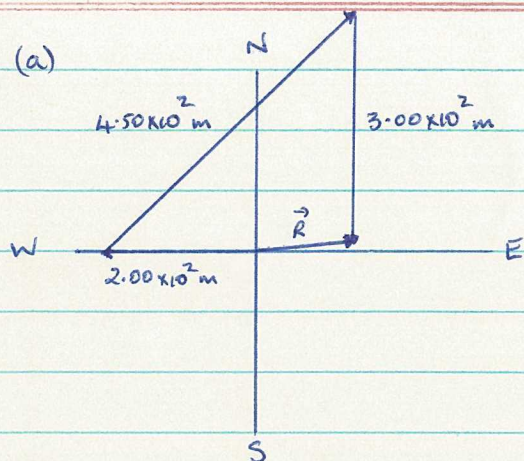
TOTAL: 13 MARKS

YEAR 10 SCIENCE

PHYSICS

TEST 2

1. (a)



$$\vec{R} = 1.2 \times 10^2 \text{ m } E 9^\circ N \quad (2)$$

$$\begin{aligned} (b) \quad v_{av} &= \frac{s}{t} \\ &= \frac{1.2 \times 10^2}{245.0} \\ &= 0.4898 \end{aligned}$$

$$\therefore \text{average velocity} = 0.49 \text{ ms}^{-1} E 9^\circ N \quad (2)$$

$$\begin{aligned} (c) \quad \text{speed} &= \frac{\text{distance}}{\text{time}} \\ &= \frac{9.50 \times 10^2}{245.0} \\ &= 3.878 \text{ ms}^{-1} \end{aligned}$$

$$\therefore \text{average speed} = 3.88 \text{ ms}^{-1} \quad (1)$$

2.

$$\begin{aligned} u &= \frac{s}{t} \\ &= \frac{1.50 \times 10^{-2}}{0.0800} \\ &= 0.188 \text{ ms}^{-1} \quad (1) \end{aligned}$$

$$\begin{aligned} v &= \frac{s}{t} \\ &= \frac{6.60 \times 10^{-2}}{0.0800} \\ &= 0.825 \text{ ms}^{-1} \quad (1) \end{aligned}$$

$$\begin{aligned} t &= 12 \times 0.0200 \\ &= 0.240 \text{ s} \quad (\frac{1}{2}) \end{aligned}$$

$$\begin{aligned} a &= \frac{v-u}{t} \\ &= \frac{0.825 - 0.188}{0.240} \\ &= 2.654 \text{ ms}^{-2} \end{aligned}$$

$$\therefore \text{acceleration} = 2.65 \text{ ms}^{-2} \quad (1\frac{1}{2})$$

3. (a) $v = 5.00 \text{ ms}^{-1}$ $a = \frac{v-u}{t}$ (forwards +ve)

$u = 20.0 \text{ ms}^{-1}$

$= \frac{5.00 - 20.0}{3.00}$

$a = ?$

$t = 3.00 \text{ s}$

$= -5.00 \text{ ms}^{-2}$

$s = ?$

$\therefore \text{acceleration} = 5.00 \text{ ms}^{-2} \text{ backwards} \quad (2)$

(b) $s = ut + \frac{1}{2} at^2$

$= (20.0)(3.00) + \frac{1}{2}(-5.00)(3.00)^2$

$= 37.5 \text{ m.}$

$\therefore \text{distance covered} = 37.5 \text{ m forwards.} \quad (2)$

TOTAL: 13